Coffee: From Cherry to Cup

Coffee is made from the seeds of a small tropical evergreen shrub. The seeds grow inside a fruit called a coffee cherry, which turns red or yellow when ripe. What people call coffee beans are therefore not true beans but the pits of these cherries, and each cherry usually holds two of them.

Two species dominate the world's coffee. Arabica is prized for its sweeter, more complex flavour and makes up the larger share of production. Robusta is hardier, contains more caffeine and has a stronger, more bitter taste. It is often used in instant coffee and in some espresso blends.

Coffee grows best in a band around the equator that is often called the coffee belt. The plants need warm temperatures, plenty of rain and no frost. Arabica generally prefers higher altitudes, where cooler air slows the ripening of the cherries and tends to give the beans a more developed flavour.

A coffee plant takes a few years to bear its first full crop after planting. The white, fragrant flowers appear after rain and last only a few days, and the cherries then need several months to ripen. Because the cherries on one branch do not ripen all at once, the best farms pick them by hand several times during a harvest.

After picking, the seeds must be separated from the fruit, and the method has a large influence on flavour. In the washed process, the fruit is removed and the seeds are fermented and rinsed before drying. In the natural process, whole cherries are dried in the sun first, which often gives the coffee a fruitier character.

The seeds are dried until they are hard and pale green, and in this state they are called green coffee. Green coffee can be stored for months without losing much quality. Roasters buy it from producers and ship it around the world in sacks.

Roasting turns green coffee brown and creates most of its aroma. Heat causes chemical changes, including the Maillard reaction between sugars and amino acids, and the beans expand and crack audibly. Light roasts keep more of the original flavour of the bean, while dark roasts taste smokier and more bitter.

Freshly roasted coffee is best ground just before brewing, because grinding exposes more surface to the air and the aroma fades quickly. The size of the grind should match the brewing method. Espresso needs a fine grind, a paper-filter pour-over needs a medium grind, and a French press works best with a coarse grind.

Brewing is a matter of extracting flavour with hot water. Espresso forces hot water through finely ground coffee under high pressure for a short time, producing a concentrated drink topped with a layer of foam called crema. A French press steeps coarse grounds in water for several minutes before a plunger holds them back.

Caffeine is the best-known compound in coffee. It is a mild stimulant that blocks receptors for adenosine, a chemical in the brain that promotes sleepiness. Many people drink coffee to feel more alert, though too much can cause jitters or interfere with sleep. Decaffeinated coffee has most of the caffeine removed before roasting.

Coffee has a long history. Legend places its discovery in Ethiopia, where the plant grows wild. Coffee drinking later became established in Yemen and then spread through the Middle East, and from there it reached Europe, where coffeehouses became popular meeting places for conversation.

Coffee is now grown in many countries, and Brazil has long been the largest producer. Growers face challenges such as leaf rust, a fungal disease, and changes in temperature and rainfall that make some areas less suitable. Many buyers pay extra for beans that are certified as grown under fair or environmentally friendly conditions.

Spent coffee grounds do not have to be thrown away. Gardeners often add them to compost, where they supply nitrogen and other nutrients to the mix. Some businesses also collect used grounds and turn them into products such as fuel pellets or ingredients for soap.

The way coffee is served varies around the world. A cup in Italy is often a short espresso drunk standing at a counter, while a Turkish cup is made by simmering very fine grounds in a small pot. Milk-based drinks such as the latte and the cappuccino combine espresso with steamed milk in different proportions.